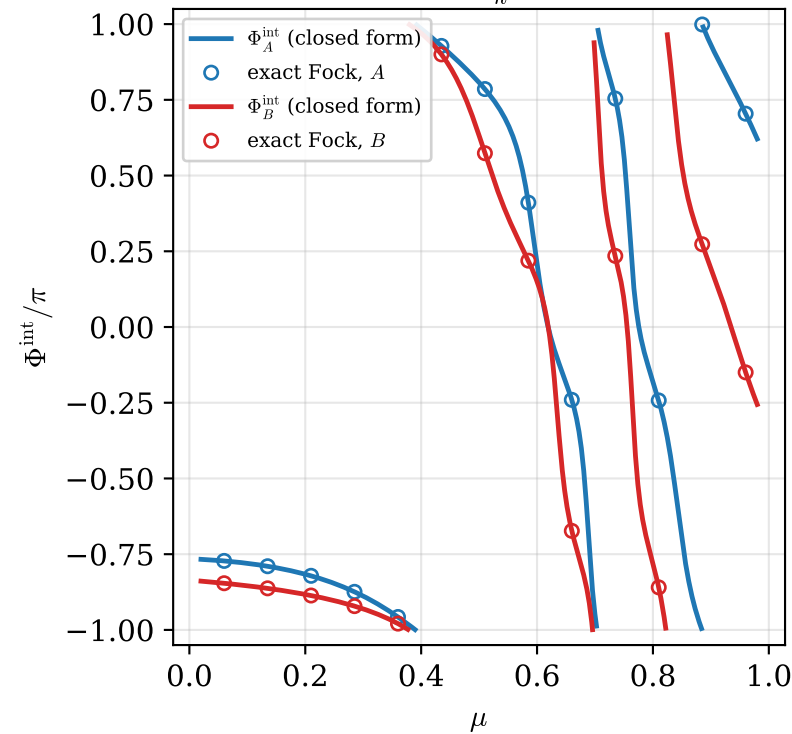


Interferometric (Sjoqvist) phase of the reduced states, branch-correlated family, $(m, n) = (0, 2)$, $\alpha = 0.5$, $\beta = 0.9$

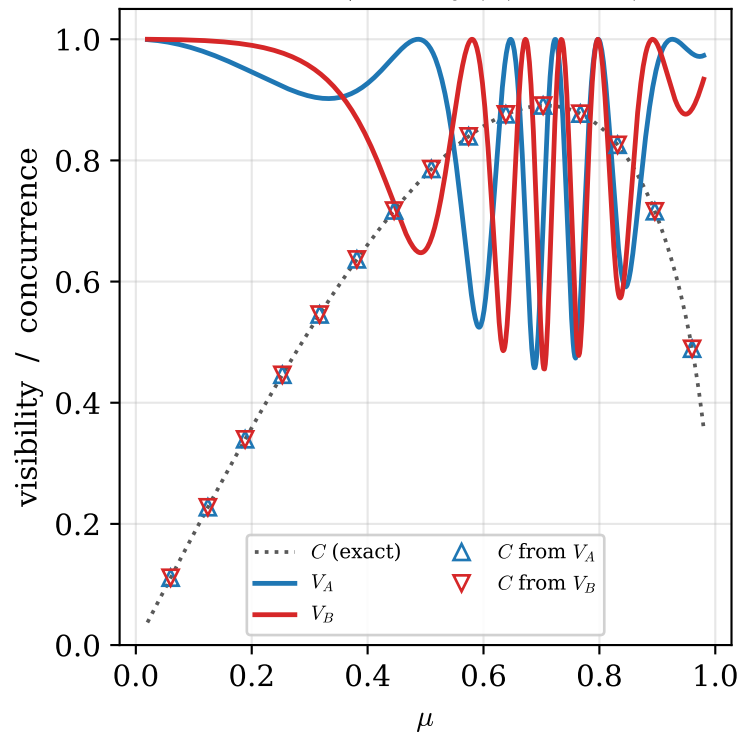
(a) laboratory frame:

$$\Phi_s^{\text{int}} = \arg \sum_k \lambda_k e^{2\pi i \nu_k}$$



(b) $V_s = \sqrt{1 - C^2 \sin^2(2\pi\delta_s)}$

$$\Rightarrow C = \sqrt{1 - V_s^2} / |\sin 2\pi\delta_s|$$



(c) simulated Mach-Zehnder fringes on mode A:

shift = $-\Phi_A^{\text{int}}$ (dotted), contrast = V_A

